NON-SCHURIAN TWISTS OF THE GROUP ASSOCIATION SCHEMES OF SUZUKI GROUPS

ABSTRACT. Let $G = \mathrm{Sz}(q)$, where $q = 2^{2m+1}$ and $m \geq 1$. We construct a Schur partition of G which has the same structure constants as its conjugacy-class partition but defines a non-Schurian association scheme. Consequently the primitive group association scheme of Suzuki group is not determined, up to combinatorial isomorphism, by its intersection numbers. This settles the Suzuki-group case of Question 8.1 of Higashitani, Kamiya, and Kurihara.

Keywords. Suzuki group; association scheme; Schur ring.

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1. INTRODUCTION

For a finite group G, its conjugacy classes define an association scheme $\mathcal{X}(G)$ on the set G: the relation associated with a conjugacy class C consists of the pairs (x, y) for which $yx^{-1} \in C$. Its intersection numbers are the structure constants of the conjugacy-class sums. The scheme $\mathcal{X}(G)$ is Schurian, because its relations are the orbitals of the action $(a, b) : x \mapsto axb^{-1}$ of $G \times G$. It is primitive when G is simple.

Higashitani, Kamiya, and Kurihara [3] construct Schur partitions with the same structure constants as the conjugacy-class partition for several finite simple groups. Their Question 8.1 asks which nonabelian finite simple groups have group association schemes that are not uniquely determined by their intersection numbers. The present paper answers that question for the whole family of Suzuki groups.

Throughout, let

$$q = 2^{2m+1}, \qquad m \geq 1, \qquad G = \mathrm{Sz}(q).$$

Choose a Borel subgroup $B = U \rtimes T$, where U is a Sylow 2-subgroup of order q^2 and T is cyclic of order $q - 1$. For $t \in T, t \neq 1$, let C_t be its G-conjugacy class. Then

$$C_t = C_{t^{-1}}, \qquad C_t \cap B = Ut \sqcup Ut^{-1}.$$

For $k \in (\mathbb{Z}/(q-1)\mathbb{Z})^\times$, define

$$D_t^{(k)} = (C_t \setminus B) \sqcup Ut^k \sqcup Ut^{-k} \qquad (t \in T \setminus \{1\}). \tag{1.1}$$

Replace each split semisimple conjugacy class C_t by $D_t^{(k)}$, and leave all other conjugacy classes unchanged. Figure 1 illustrates the two parts of $D_t^{(k)}$.

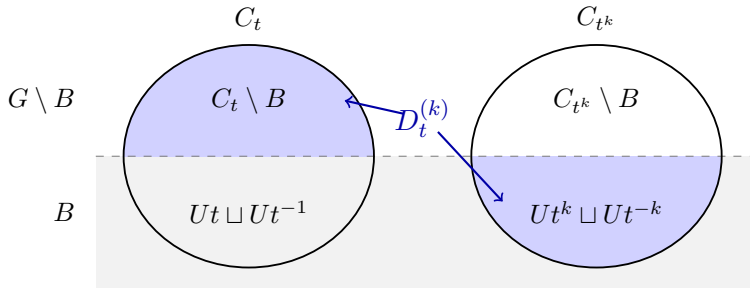


FIGURE 1. The two highlighted regions form the switched basic set $D_t^{(k)} = (C_t \setminus B) \sqcup (C_{t^k} \cap B)$.

Denote the resulting partition by $\mathcal{D}_k$. Our main result is as follows.

Theorem 1.1. *For every $k \in (\mathbb{Z}/(q-1)\mathbb{Z})^\times$, the partition $\mathcal{D}_k$ is a Schur partition of G, and the correspondence from each conjugacy class to its replacement preserves all structure constants. If $k \not\equiv \pm 1 \pmod{q-1}$, the associated Cayley scheme is non-Schurian and therefore is not combinatorially isomorphic to $\mathcal{X}(G)$. If $k \equiv \pm 1 \pmod{q-1}$, then $\mathcal{D}_k = \mathrm{Cl}(G)$.*

Since $q - 1 \geq 7$ is odd, $k = 2$ always satisfies the nontriviality condition.

The switching in (1.1) is the analogue of the Borel switching construction in [3, Section 4]. Two arguments establish the Suzuki-group case. First, all sets gUt, with $g \notin B$ and $t \in T$, are conjugate as subsets of G. Second, a local graph defined using the involution conjugacy class detects non-Schurity. Its cliques of size q are cosets of centers of Sylow 2-subgroups, and a trace calculation controls which of them survive the switching.

2. Preliminaries

2.1. **Schur partitions and local graphs.** For a subset A of a finite group, put $\underline{A} = \sum_{a \in A} a \in \mathbb{C}G$. A partition $\mathcal{S}$ of G is a *Schur partition* if it contains $\{1\}$, is closed under inversion, and its part sums span a subalgebra of $\mathbb{C}G$. Thus, for $A, B \in \mathcal{S}$,

$$\underline{AB} = \sum_{C \in \mathcal{S}} c_{AB}^C \underline{C}.$$

The corresponding Cayley association scheme has relations

$$R_A = \{(x, y) \in G^2 : yx^{-1} \in A\} \qquad (A \in \mathcal{S}).$$

A bijection of parts preserving the constants c_{AB}^C is an *algebraic isomorphism*. A *combinatorial isomorphism* is a bijection of the underlying point sets that maps each relation onto a relation. *Separability* requires every algebraic isomorphism to be induced by a combinatorial one.

We use the local graph notion of [3, Definition 3.7]. An element of $\mathcal{S}$ is called a *basic set*. Let $X \in \mathcal{S}$ be a basic set, let Y be a union of basic sets not containing 1, and let $g \in G$. The *local graph* $\Gamma_Y(X, g)$ has vertex set $R_X(g) = Xg$, with an arc from a to b when $ba^{-1} \in Y$. We abbreviate

$$\Gamma_Y(X) := \Gamma_Y(X, 1).$$

If $Y = Y^{-1}$, we regard this as an undirected graph.

Lemma 2.1. *If the association scheme corresponding to $\mathcal{S}$ is Schurian, then every local directed graph $\Gamma_Y(X, g)$ is vertex-transitive.*

Proof. Suppose the relations are the orbitals of a transitive permutation group K. The stabilizer K_g is transitive on the relation neighborhood $R_X(g) = Xg$ and preserves every relation, hence also the adjacency relation defined by Y. □

In the proof of Theorem 1.1, we exhibit a local graph on a switched basic set that is not vertex-transitive, thereby proving that the switched scheme is non-Schurian.

Throughout this paper, let

$$\mathcal{I} = \{g \in G \setminus \{1\} : g^2 = 1\}$$

be the set of involutions of G. As in [3, Section 4.2], we use Δ for the Cayley graph on the whole set G in which x and y are adjacent when $yx^{-1} \in \mathcal{I}$.

2.2. **Suzuki groups.** We include the matrix realization and Bruhat decomposition of the Suzuki groups, which are standard results; see Suzuki [4], Bäärnhielm [1, Section 2.1.1], and Breuillard–Green–Tao [2, Definition 3.1]. Note that [2] uses the equivalent parameter $u(a, b + a^{\theta+1})$ for its unipotent matrices.

Set $\theta = 2^{m+1}$, so that $a^{\theta^2} = a^2$ for $a \in \mathbb{F}_q$. We use the standard matrix realization of the Suzuki group. For $a, b \in \mathbb{F}_q$ and $c \in \mathbb{F}_q^\times$, put

$$u(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ b & a^\theta & 1 & 0 \\ a^{\theta+2} + ab + b^\theta & a^{\theta+1} + b & a & 1 \end{pmatrix}, \tag{2.1}$$

$$h(c) = \operatorname{diag}(c^{\theta+1}, c, c^{-1}, c^{-\theta-1}), \qquad w = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{2.2}$$

Then $U = \{u(a,b) : a, b \in \mathbb{F}_q\}$ and $T = \{h(c) : c \in \mathbb{F}_q^\times\}$. The standard structure theory gives

$$G = B \sqcup UTwU, \qquad B = U \rtimes T, \qquad wtw = t^{-1} \quad (t \in T), \tag{2.3}$$

with unique parameters in each of the two cells. In particular,

$$|G| = q^2(q^2 + 1)(q - 1). \tag{2.4}$$

Let $\Omega = G/B = \{gB : g \in G\}$. We regard each left coset as a point, and let G act on these points by left multiplication. We name two particular points

$$\infty := B, \qquad 0 := wB.$$

For a point $\alpha = gB$, let $B_\alpha = G_\alpha$ be its stabilizer. Explicitly,

$$B_\alpha = \{h \in G : hgB = gB\} = gBg^{-1}.$$

We also define

$$U_\alpha := gUg^{-1}, \qquad Z_\alpha := Z(U_\alpha) = gZ(U)g^{-1}.$$

Thus the three subgroups B_α, U_α, and Z_α are obtained from B, U, and its center $Z(U)$ by the same conjugation. In particular, U_α is a normal Sylow 2-subgroup of B_α. These definitions do not depend on the representative g of the coset: every other representative has the form gb with $b \in B$, and $U \lhd B$ gives

$$(gb)U(gb)^{-1} = gUg^{-1}.$$

The same conclusion holds for B_α and Z_α. For the two distinguished points ∞ and 0, the definitions give

$$\begin{aligned} B_\infty &= B, & U_\infty &= U, & Z_\infty &= Z(U), \\ B_0 &= wBw^{-1}, & U_0 &= wUw^{-1}, & Z_0 &= wZ(U)w^{-1}. \end{aligned}$$

We call an element $x \in G$ a *split semisimple element* if $x \in C_t$ for some $t \in T \setminus \{1\}$. We record the consequences of this action used below.

Lemma 2.2. *The following assertions hold.*

(i) *The subgroup U_α acts regularly on $\Omega \setminus \{\alpha\}$. The action of G on Ω is doubly transitive of degree $q^2 + 1$. Every nonidentity element of G fixes at most two points.*

(ii) *Distinct U_α intersect trivially. The involutions of U_α are exactly $Z_\alpha \setminus \{1\}$, and $|Z_\alpha| = q$. The set $\mathcal{I}$ of all involutions of G is a single conjugacy class.*

(iii) *If $t \in T \setminus \{1\}$, then Ut is a single U-conjugacy orbit. Moreover,*

$$C_G(t) = T, \quad C_t \cap B = Ut \sqcup Ut^{-1}, \quad |C_t| = q^2(q^2+1). \tag{2.5}$$

The equality $C_s = C_t$, for $s, t \in T \setminus \{1\}$, holds exactly when $s \in \{t, t^{-1}\}$.

(iv) *If a split semisimple element x fixes α, then $xU_\alpha \subseteq x^G$. In particular, $xZ_\alpha \subseteq x^G$.*

Proof. Direct multiplication in (2.1) gives

$$u(a,b)u(c,d) = u(a+c, b+d+a^\theta c), \tag{2.6}$$

$$u(a,b)^2 = u(0, a^{\theta+1}), \tag{2.7}$$

$$h(c)^{-1}u(a,b)h(c) = u(c^\theta a, c^{\theta+2}b). \tag{2.8}$$

The integers $\theta = 2^{m+1}$ and $\theta + 2 = 2(2^m+1)$ are coprime to $q-1$. For the latter assertion, a common divisor of 2^m+1 and $2^{2m+1}-1$ divides 1, and $q-1$ is odd. If $t = h(c) \neq 1$, then $c \neq 1$, so $c^\theta \neq 1$ and $c^{\theta+2} \neq 1$. By (2.8), the equality $t^{-1}u(a,b)t = u(a,b)$ therefore forces $a = b = 0$. Consequently,

$$C_U(t) = \{1\} \qquad (t \in T \setminus \{1\}). \tag{2.9}$$

(i) *Regular action.* The number of points in the coset action is

$$|\Omega| = [G : B] = \frac{q^2(q^2+1)(q-1)}{q^2(q-1)} = q^2+1.$$

Since T normalizes U and $tw = wt^{-1}$, the Bruhat decomposition (2.3) can be written as $G = B \sqcup UwB$. Hence every point other than ∞ has the form uwB with $u \in U$. There are q^2 such points and $|U| = q^2$, so their representatives u are unique. Under the identification $u \leftrightarrow uwB$, U acts by left multiplication on itself, so the action is regular. For $\alpha = gB$, conjugate by g to obtain the regular action of $U_\alpha = gUg^{-1}$ on $\Omega \setminus \{\alpha\}$. In particular, every nonidentity element of U_α fixes exactly the point α.

Double transitivity. The action of G on its cosets is transitive. Its stabilizer $G_\infty = B$ contains U, which is transitive on $\Omega \setminus \{\infty\}$. Thus G is doubly transitive on Ω.

The bound on fixed points. The group T fixes $0 = wB$. Since $B = U \rtimes T$ and no nonidentity element of U fixes 0, the pointwise stabilizer of ∞ and 0 is

$$G_{\infty,0} = T.$$

For $t \in T$ and $u \in U$,

$$t \cdot (uwB) = (tut^{-1})wB.$$

By uniqueness of the labels u, the point uwB is fixed by t exactly when $tut^{-1} = u$. If $t \neq 1$, (2.9) forces $u = 1$, corresponding to the point 0. Thus a nonidentity element of T fixes precisely ∞ and 0. Now suppose a nonidentity element of G fixed three distinct points. By double transitivity, conjugate this element so that two of its fixed points become ∞ and 0. The conjugate is a nonidentity element of T and still fixes a third point, a contradiction. This completes the proof of (i).

(ii) If $\alpha \neq \beta$, a nonidentity element in $U_\alpha \cap U_\beta$ would fix both points, although a nonidentity element of U_α fixes only α. Hence $U_\alpha \cap U_\beta = \{1\}$.

Equations (2.6)–(2.7) show that the involutions in U are exactly $u(0,b)$ with $b \neq 0$, and that

$$Z(U) = \{u(0,b) : b \in \mathbb{F}_q\}.$$

Indeed, an element with $a \neq 0$ cannot commute with every $u(c,0)$: this would require $a^\theta c = c^\theta a$ for all $c \in \mathbb{F}_q$.

Equation (2.8) shows that T acts transitively by conjugation on the involutions in U. Every involution of G lies in a Sylow 2-subgroup, and every such subgroup is conjugate to U. Hence every involution of G is conjugate to an involution in U, so all involutions of G belong to a single conjugacy class. This proves (ii).

(iii) For $t \in T \setminus \{1\}$, the orbit–stabilizer theorem and (2.9) give

$$|\{utu^{-1} : u \in U\}| = |U : C_U(t)| = |U|.$$

This orbit is contained in Ut, by normality of U in B, and hence equals Ut.

In particular, every element of $B \setminus U$ is U-conjugate to its image in T. As proved above, the two fixed points of t are ∞ and 0. Their pointwise stabilizer is T, and their setwise stabilizer is $T \sqcup Tw$, because w interchanges them. Every element of Tw inverts t and therefore cannot centralize t, since t has odd order. Thus $C_G(t) = T$. Likewise, any conjugation carrying one nonidentity element of T to another preserves this pair. Thus two elements $s, t \in T \setminus \{1\}$ are conjugate in G if and only if $s \in \{t, t^{-1}\}$. No nonidentity element of U is conjugate to t, since the former has 2-power order and the latter has odd order. Formula (2.5) and (iii) follow.

(iv) Choose $g \in G$ with $\alpha = gB$ and put $y = g^{-1}xg$. Then $y \in B$, and its nontrivial odd order implies $y \notin U$. Write $y = vt$ with $v \in U$ and $t \in T \setminus \{1\}$. Since $U \lhd B$, we have $yU = vtU = Ut$, which by (iii) is a single U-conjugacy orbit containing y. Hence $yU \subseteq y^G$. Conjugating by g gives $xU_\alpha = g(yU)g^{-1} \subseteq y^G = x^G$. □

3. Algebraic isomorphism

3.1. **A uniform coset distribution.** The next observation explains the algebraic part of the construction. It uses only (2.3) and the odd order of T.

Lemma 3.1. *For every $g \in G \setminus B$ and $t \in T$, the subset gUt is conjugate to wU. Consequently, for every conjugacy class C,*

$$|gUt \cap C| = |wU \cap C|. \tag{3.1}$$

Proof. Write $g = u_1hwu_2$ with $u_1, u_2 \in U$ and $h \in T$. Normality of U in B gives

$$u_1^{-1}(gUt)u_1 = hwUt = ht^{-1}wU.$$

Conjugating the last set by $a \in T$ gives

$$a(ht^{-1}wU)a^{-1} = ht^{-1}a^2wU.$$

Since $|T|$ is odd, the squaring map on T is bijective. We can choose a with $a^2 = th^{-1}$. This proves that gUt is conjugate to wU. □

For conjugacy classes C, E, define, for $z \in B$,

$$F_{C,E}(z) = |\{(x, y) \in (C \setminus B) \times (E \setminus B) : xy = z\}|.$$

Lemma 3.2. *The function $F_{C,E}$ is constant on $B \setminus U$.*

Proof. Recall $B \setminus U = \sqcup_{t \neq 1} Ut$. For each $t \neq 1$, conjugation by U is transitive on Ut and preserves $C \setminus B$ and $E \setminus B$. Thus $F_{C,E}$ has a constant value f_t on Ut. Summing over $z \in Ut$ yields

$$q^2 f_t = \sum_{x \in C \setminus B} |x^{-1}Ut \cap E|.$$

Here $x^{-1}Ut \subseteq G \setminus B$, so there is no need to remove B from E in the summand. Lemma 3.1 makes the right-hand side independent of t. □

3.2. **The counting argument.** Write $\sigma(t) = t^k$. For every conjugacy class C, denote its switched counterpart in $\mathcal{D}_k$ by D_C. Its intersection with $G \setminus B$ is unchanged. If $C = C_t$, its intersection with B is changed from $Ut \sqcup Ut^{-1}$ to $U\sigma(t) \sqcup U\sigma(t)^{-1}$. Every other class meets B in a subset of U, possibly empty, and is unchanged.

Proposition 3.3. *For conjugacy classes C, E, L and every $z \in D_L$,*

$$|\{(x, y) \in D_C \times D_E : xy = z\}| = c_{CE}^L, \tag{3.2}$$

where c_{CE}^L is the corresponding intersection number of $\mathcal{X}(G)$. In particular, $C \mapsto D_C$ is an algebraic isomorphism.

Proof. The sets D_C partition G because σ permutes the nonidentity elements of T and respects inversion. Their sizes are unchanged, the identity singleton is unchanged, and $D_{C^{-1}} = (D_C)^{-1}$. It remains to prove (3.2).

First suppose $z \notin B$. In a factorization $xy = z$, it cannot happen that $x, y \in B$. The contribution with $x, y \notin B$ is unchanged. The contribution from pairs with $x \in A \subseteq B$ and $y \in E \setminus B$ is

$$|\{a \in A : z^{-1}a \in E^{-1}\}|.$$

Replacing $A = Ut \sqcup Ut^{-1}$ by $U\sigma(t) \sqcup U\sigma(t)^{-1}$ preserves this count, by Lemma 3.1. For the case with $x \notin B$ and $y \in B$, with second factor $b \in B$, the condition $zb^{-1} \in C$ is equivalent to $bz^{-1} \in C^{-1}$, and hence to $z^{-1}b \in C^{-1}$ by z^{-1}-conjugation. Lemma 3.1 applies again. All contributions therefore agree with the original ones at z, which belongs to L.

Now let $z \in B$. A factorization has either both factors in B or both outside B. If $z \in U$, take $z_0 = z$. If $z \in Uv$ with $v \neq 1$, choose any $z_0 \in U\sigma^{-1}(v)$. In both cases $z_0 \in L$. The contribution of two outside factors at z after switching equals the original contribution at z_0: for $z \in U$ this is immediate, and for $z \notin U$ it follows from Lemma 3.2.

For the contribution of two factors in B, use the quotient homomorphism $B \longrightarrow T$. For $s, t, v \in T$ and $z \in Uv$,

$$|\{(x, y) \in Us \times Ut : xy = z\}| = \begin{cases} q^2, & st = v, \\ 0, & st \neq v. \end{cases} \tag{3.3}$$

Also, for any $A \subseteq U$,

$$\begin{aligned} |\{(x, y) \in A \times Ut : xy = z\}| &= |\{(x, y) \in Ut \times A : xy = z\}| \\ &= \begin{cases} |A|, & t = v, \\ 0, & t \neq v. \end{cases} \end{aligned} \tag{3.4}$$

These identities follow by choosing the factor in the specified subset and solving uniquely for the other factor. Products of two unchanged subsets of U are supported on U and remain unchanged there. Equations (3.3)–(3.4) handle all other cases, since σ preserves the multiplication equations in T. Thus the contribution inside B at z after switching agrees with the original one at z_0.

Adding the contributions gives c_{CE}^L. This proves (3.2), the Schur-partition assertion, and the algebraic isomorphism. □

4. Involution cliques and a trace bound

Recall that $\Delta = \Gamma_{\mathcal{I}}(G)$ is the involution graph on G and that the local graph on a basic set X is $\Gamma_{\mathcal{I}}(X) = \Delta[X]$. The graph Δ is undirected because involutions are self-inverse. Also recall that $Z_\alpha = Z(U_\alpha)$ has order q.

Lemma 4.1. *The maximum size of a clique in Δ is q. For any $x \in G$, the cliques of size q containing x are precisely*

$$xZ_\alpha \qquad (\alpha \in \Omega).$$

These cliques are distinct for distinct α.

Proof. Since $\mathcal{I}$ is invariant under conjugation, left translations are automorphisms of Δ: for $x, a, b \in G$,

$$(x^{-1}b)(x^{-1}a)^{-1} = x^{-1}(ba^{-1})x \in \mathcal{I} \quad \Longleftrightarrow \quad ba^{-1} \in \mathcal{I}.$$

Let K be a clique containing x. Its left translate $x^{-1}K$ is a clique containing 1, so all its other vertices are involutions. For any two distinct such vertices a, b, adjacency gives $ba^{-1} = ba \in \mathcal{I}$, and hence $ba = (ba)^{-1} = ab$. These vertices therefore generate an elementary abelian 2-subgroup, contained in some Sylow 2-subgroup U_α. By Lemma 2.2(ii), $x^{-1}K \subseteq Z_\alpha$. Thus $|K| \leq q$, and equality forces $x^{-1}K = Z_\alpha$, or equivalently $K = xZ_\alpha$.

Conversely, each Z_α is a clique of size q, since its nonidentity elements are involutions. Left translation shows that xZ_α is also a clique of size q containing x. Lemma 2.2(ii) implies that distinct Z_α intersect trivially, so the cosets xZ_α are distinct as α varies. □

Lemma 4.2. *If x does not fix $\alpha \in \Omega$, then for every conjugacy class C of G,*

$$|xZ_\alpha \cap C| \leq 2. \tag{4.1}$$

Proof. By simultaneous conjugation, assume $\alpha = \infty$, so $x \notin B$. Put $z(b) = u(0, b)$. From (2.1),

$$z(b) = I + b(E_{31} + E_{42}) + b^\theta E_{41},$$

where E_{ij} are matrix units. Writing $x = (x_{ij})$, we obtain

$$\operatorname{tr}(xz(b)) = \operatorname{tr}(x) + (x_{13} + x_{24})b + x_{14}b^\theta. \tag{4.2}$$

By the Bruhat decomposition, $x = u_1 h w u_2$, so that $x_{14} = h_{11} \neq 0$.

For $A \in \mathbb{F}_q$ and $B_0 \in \mathbb{F}_q^\times$, each fiber of $b \mapsto Ab + B_0 b^\theta$ has at most two elements. Indeed, this map is additive, and a nonzero element d of its kernel satisfies

$$d^{\theta-1} = A/B_0.$$

Recall the identity $\gcd(2^r - 1, 2^s - 1) = 2^{\gcd(r,s)} - 1$ for positive integers r, s, which follows from the Euclidean algorithm applied to the exponents. There is at most one such d, because

$$\gcd(\theta - 1, q - 1) = 2^{\gcd(m+1, 2m+1)} - 1 = 1,$$

the map $d \mapsto d^{\theta-1}$ is a bijection of $\mathbb{F}_q^\times$. The trace is constant on C, so (4.2) implies (4.1). □

Corollary 4.3. *Every vertex x of the local graph $\Gamma_{\mathcal{I}}(C_t)$, with $t \neq 1$, lies in exactly two cliques of size q. They are xZ_α for the two fixed points α of that vertex x.*

Proof. If x fixes α, then $xZ_\alpha \subseteq C_t$ by Lemma 2.2(iv). If it does not, Lemma 4.2 excludes this inclusion, since $q > 2$. Now use Lemma 4.1 and the fact that a split semisimple element has exactly two fixed points. □

5. A local obstruction to Schurity

Assume $k \not\equiv \pm 1 \pmod{q-1}$. Choose $t \in T$ such that

$$t^k \notin \{t, t^{-1}\}; \tag{5.1}$$

a generator of T has this property. Abbreviate

$$C = C_t, \qquad C' = C_{t^k}, \qquad D = D_t^{(k)}.$$

The classes C and C' are distinct, and

$$D \setminus B = C \setminus B, \qquad D \cap B = C' \cap B = Ut^k \sqcup Ut^{-k}, \qquad D \cap C \cap B = \varnothing. \tag{5.2}$$

Figure 2 illustrates the two parts of D.

The involution class $\mathcal{I}$ is unchanged by switching and is a basic set of both partitions. Thus $\Gamma_{\mathcal{I}}(C)$ and $\Gamma_{\mathcal{I}}(D)$ are the local graphs on C and D in the original and switched schemes.

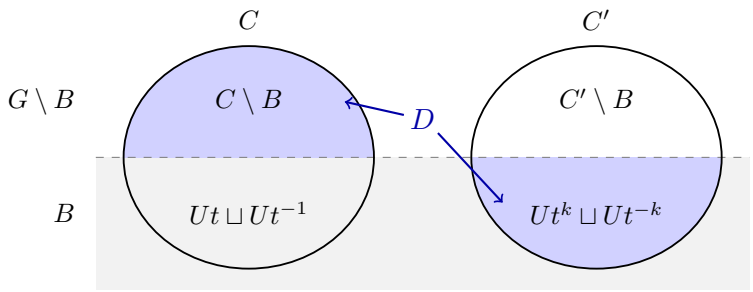


FIGURE 2. The two highlighted regions form the switched basic set $D = (C \setminus B) \sqcup (C' \cap B)$.

Lemma 5.1. *If $\alpha \neq \infty$, every left coset of Z_α meets B in at most one element.*

Proof. Every nonidentity element of Z_α fixes only α, and therefore $Z_\alpha \cap B = \{1\}$. If $xz_1, xz_2 \in B$ with $z_1, z_2 \in Z_\alpha$, then $(xz_1)^{-1}(xz_2) = z_1^{-1}z_2 \in Z_\alpha \cap B$, whence $z_1 = z_2$. □

Lemma 5.2. *If $x \in D$ and $xZ_\alpha \subseteq D$, then x fixes α.*

Proof. Suppose otherwise. If $\alpha = \infty$, then $x \notin B$ and the entire coset xZ_∞ is outside B. By (5.2) it would be contained in C, contrary to Lemma 4.2. If $\alpha \neq \infty$, Lemma 5.1 leaves at least $q - 1$ elements of this coset outside B. All would have to belong to C, again contradicting Lemma 4.2, because $q - 1 > 2$. □

Lemma 5.3. *Every vertex of $D \cap B$ lies in exactly one clique of size q in $\Gamma_{\mathcal{I}}(D)$.*

Proof. Let $x \in D \cap B \subseteq C'$. Its two fixed points are ∞ and a point $\beta \neq \infty$. By Lemmas 4.1 and 5.2, the only possible cliques are xZ_∞ and xZ_β. The first lies in the same U-coset as x, and hence in D, by 2.2(iii).

For the second, Lemma 2.2(iv) gives $xZ_\beta \subseteq C'$. It meets B only at x, by Lemma 5.1. Its other $q - 1$ vertices of this clique are in $C' \setminus B$, not contained in D. □

Lemma 5.4. *At least $q^2(q-1)^2$ vertices of $D \setminus B$ lie in exactly two cliques of size q in $\Gamma_{\mathcal{I}}(D)$.*

Proof. By Corollary 4.3, every vertex of $C \setminus B$ lies in exactly two cliques of size q in $\Gamma_{\mathcal{I}}(C)$. Such a clique survives in $\Gamma_{\mathcal{I}}(D)$ whenever it is disjoint from B. Conversely, for any clique xZ_α of size q in $\Gamma_{\mathcal{I}}(D)$ through $x \in C \setminus B$, x fixed α, by Lemma 5.2. Then the entire coset xZ_α lies in C.

Count the original cliques that contain points both in B and outside B. For each $r \in C \cap B$, its two original cliques are rZ_∞, which is contained in B, and rZ_β, where β is its other fixed point. The latter meets B only at r, by Lemma 5.1. Consequently these crossing cliques are indexed without repetition by the elements of $C \cap B$. There are $2q^2$ of them, each with $q - 1$ vertices outside B.

Thus at most $2q^2(q-1)$ vertices outside B lie in a crossing clique. On the other hand,

$$|C \setminus B| = q^2(q^2+1) - 2q^2 = q^2(q^2-1).$$

At least

$$q^2(q^2-1) - 2q^2(q-1) = q^2(q-1)^2$$

vertices lie in no crossing clique and therefore retain both of their original cliques. No additional cliques are possible by the first paragraph of the proof. □

Proof of Theorem 1.1. Proposition 3.3 proves the Schur-partition and algebraic-isomorphism assertions. Under $k \not\equiv \pm 1 \pmod{q-1}$, Lemmas 5.3 and 5.4 give vertices of $\Gamma_{\mathcal{I}}(D)$ lying in different numbers of cliques of size q. Since this number is invariant under graph automorphisms, $\Gamma_{\mathcal{I}}(D)$ is not vertex-transitive. Lemma 2.1 proves that the switched scheme is non-Schurian. It cannot be combinatorially isomorphic to the Schurian scheme $\mathcal{X}(G)$. Finally, if $k \equiv \pm 1 \pmod{q-1}$, then $\mathcal{D}_k = \mathrm{Cl}(G)$, so the construction recovers $\mathcal{X}(G)$. □

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